The CTF challenge this writeup is for is the Faulty image 4

Firstly, we are given an image with the flag “LITCTF{fau1ty_f0uR_12ox!}” encrypted on it. However, when we first enter the image, the correct flag is not shown.

There is also a tip saying that certain pixels have unusually small blue values that are clipped to 4 bits. We are asked to identify these pixels. Furthermore, the question tells us that the encoding information is scattered around the different pixels. These pixels are unusually small blue value pixels, and they hold 4 bits of data, or nibbles.

Since each letter requires 8 bits of data, the information is hidden in the consecutive pixels using the blue channel. We need to run a script to extract the blue value of the pixel containing extremely low blue value and piece them together to decode the flag.

The code and image are in the same folder as this writeup. The flag is LITCTF{pr1m3_n1bbl3s_77zq!}.